SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
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COURSE OUTCOME COVERED

CO4:
Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

Annexure A

Industry Problem-Solving Task

1. A star topology is ideal for a 20-person startup because all devices connect to a central switch, making the network easy to manage and troubleshoot. At the core sits a Layer 2/Layer 3 managed switch (such as a Cisco Catalyst or Netgear ProSafe) with at least 24 ports to accommodate all workstations, printers, and VoIP phones with room to grow. Each device connects to the switch using Cat6 UTP cables, which support Gigabit speeds up to 100 meters — more than sufficient for any typical office floor plan. The switch uplinks to a router/firewall (such as a Cisco RV series or pfSense appliance) that handles internet access, DHCP, and network address translation. A Wi-Fi access point connected to the

switch extends wireless coverage for laptops and mobile devices. The key advantage of this design is fault isolation: if one workstation's cable fails, only that single node goes offline, leaving all other 19 employees unaffected. Centralised cable management also simplifies future moves, adds, and changes.

2. A bank demands near-zero downtime, making a full or partial mesh topology the right choice. In a full mesh, every core network device — routers, core switches, and servers — connects directly to every other device, so no single link or node failure can bring down the network. For a bank, a partial mesh is typically deployed at the core layer (connecting all core switches and firewalls to each other) while the distribution and access layers use redundant uplinks to at least two core switches. Dual enterprise-grade routers (such as Cisco ASR or Juniper MX series) provide BGP-based redundant internet connectivity. All links between core devices use fiber optic cables — either single-mode for long distances between buildings or multi-mode OM4 within the data center — because fiber is immune to electromagnetic interference, supports very high bandwidth, and is essential for maintaining low-latency financial transactions. Redundant power supplies, UPS systems, and RAID storage arrays further complement the mesh design. The justification is straightforward: when a link fails in a mesh, traffic automatically reroutes through alternative paths in milliseconds using protocols like OSPF or BGP, satisfying regulatory requirements for uptime and data integrity.

3. **3.** For a small classroom lab with a tight budget, bus topology offers the lowest hardware cost because all computers share a single common communication line — the bus — eliminating the need for a central switch or hub. A single coaxial cable (RG-58, 50-ohm) or, in a modern variation, a single Cat5e/Cat6 cable daisy-chained through an inexpensive unmanaged hub runs along the length of the room. Each computer connects to the backbone using a T-connector and a BNC connector in a classic coaxial setup, with terminators (50-ohm) placed at both ends of the cable to absorb signals and prevent reflection. For a more contemporary low-cost implementation, a single inexpensive 8- or 16-port unmanaged switch replaces the terminators and uses Cat5e UTP patch cables to each workstation — achieving bus-like simplicity at minimal cost. The primary limitation is that the network is susceptible to collisions and a single cable break disrupts all connected machines, but for a supervised classroom environment with light usage (internet browsing, file sharing), this trade-off is entirely acceptable. No dedicated server is required; a teacher's PC can share resources using built-in Windows/Linux file sharing.

4. A large university campus serves thousands of users across multiple buildings — lecture halls, labs, libraries, dormitories, and administration — making a single topology inadequate. A hybrid topology combines the best elements of multiple designs: a fiber-based ring or mesh at the campus backbone level for redundancy, a star topology within each building for manageability, and bus or tree structures within individual labs for cost-efficiency. At the campus core, high-speed fiber optic cables (single-mode, 10 Gbps or 40 Gbps) interconnect core routers and multilayer switches using a ring or partial mesh arrangement, ensuring that no single link failure isolates an entire building. Each building has its own distribution switch connected to the campus core via dual fiber uplinks. Within each building, a traditional star topology branches from the distribution switch to floor-level access switches, which then connect individual devices via Cat6 cables. This hierarchical three-layer model — core, distribution, and access — is the industry standard for enterprise and campus networks because adding a new building simply means installing a new distribution switch and running fiber to the core, with no disruption to existing infrastructure. Wireless access points throughout the campus connect via PoE (Power over Ethernet) to access switches, eliminating the need for separate power cabling. The hybrid design thus delivers redundancy at the backbone, simplicity at the building level, and scalability campus-wide.

5. Designing a network for a company with multiple departments involves creating a structured and scalable architecture using switches and routers to ensure efficient communication and security. Each department (such as HR, Finance, Sales, and IT) is connected through dedicated Layer 2 switches that form the access layer, allowing devices like computers and printers to communicate within the same department. These switches are then

connected to a central Layer 3 switch or router, which acts as the core layer and enables communication between different departments through inter-VLAN routing. Virtual LANs (VLANs) are configured to logically separate each department's network, improving security and reducing broadcast traffic. A router is also used to connect the internal network to external networks such as the internet, often with firewall functionality for security. Redundant links and backup paths can be included to ensure reliability, while proper IP addressing and subnetting are applied for efficient network management. This hierarchical design improves performance, enhances security, and allows easy expansion as the company grows.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Problem Analysis	15%	Clearly identifies and deeply analyzes the industry problem with all factors	Good understanding with minor gaps	Basic understanding; some key aspects missing	Poor or incorrect understanding of the problem
Solution Design	20%	Innovative, practical, and industry-relevant solution	Effective solution with minor limitations	Partially suitable solution	Inappropriate or unclear solution
Technical Implementation	20%	Fully functional, accurate, and well-executed implementation	Mostly functional with minor errors	Partially implemented solution	Incomplete or non-functional implementation
Use of Tools & Technologies	10%	Appropriate and advanced use of relevant tools and technologies	Correct use of tools	Limited or basic use of tools	Incorrect or no use of tools
Problem-Solving Approach	10%	Logical, systematic, and well-structured approach	Mostly logical approach	Somewhat disorganized approach	No clear approach
Innovation & Creativity	10%	Highly creative and original solution	Some creativity	Limited originality	No creativity
Testing & Validation	5%	Thorough testing with real-world scenarios	Adequate testing	Minimal testing	No testing
Documentation & Presentation	5%	Clear, professional, and well-documented	Good documentation	Basic documentation	Poor or no documentation
Teamwork / Collaboration	5%	Excellent coordination and contribution (if team-based)	Good collaboration	Limited participation	No collaboration or contribution

ANSWERS

Question 1:Design a Network for a 20-Person Startup Using Star Topology

Problem Analysis:

A startup with 20 employees requires a network that is reliable, easy to manage, scalable, and cost-effective. A **Star Topology** is the best choice because every device is connected to a central switch, making troubleshooting and maintenance simple.

Solution Design

Topology Used
Star Topology
Networking Devices

Device	Specification	Purpose
Layer 2 Managed Switch	24-Port Cisco Catalyst/Netgear ProSafe	Connects all devices
Router/Firewall	Cisco RV Series / pfSense	Internet access, DHCP, NAT
Wi-Fi Access Point	Wi-Fi 6 Access Point	Wireless connectivity
Workstations	20 PCs/Laptops	User devices
Printer	Network Printer	Shared printing

Transmission Media:

- **Cat6 UTP Cable**
- Speed: **1 Gbps**
- Maximum Length: **100 meters**
- RJ-45 Connector

Network Diagram:

```
              Internet
                 |
              Router/Firewall
                 |
            24-Port Switch
       -------------------------------
       ||||||||||||||||
     PC1 PC2 PC3 ... PC20 Printer Wi-Fi
```

Working:

1. Every computer connects directly to the **central switch**.
2. The switch forwards data only to the intended destination.
3. The router provides internet access.
4. The Wi-Fi access point provides wireless connectivity.
5. If one cable fails, only that device is disconnected.

Advantages:

- Easy installation
- Simple troubleshooting
- High performance
- Easy expansion
- Better security
- Reliable communication

Justification:

A **Star Topology** is ideal for a 20-person startup because it provides centralized management, high-speed communication, easy fault isolation, and supports future expansion. Cat6 UTP cables ensure Gigabit Ethernet performance up to 100 meters, making the network efficient and reliable for office use.

Conclusion:
A Star Topology using a **24-port managed switch**, **Cat6 UTP cables**, **router/firewall**, and **Wi-Fi access point** offers a scalable, secure, and reliable network solution for a 20-person startup.

Question 2: Design a Reliable Network for a Bank Using Mesh Topology

Problem Analysis:

A bank requires a network with **near-zero downtime**, high security, and continuous connectivity. A **Partial Mesh Topology** is the best choice because it provides multiple communication paths, ensuring that if one link fails, data is automatically rerouted through another path.

Solution Design:

Topology Used
Partial Mesh Topology
Networking Devices

Device	**Specification**	**Purpose**
Core Routers	Cisco ASR / Juniper MX	Internet connectivity
Core Switches	Layer 3 Managed Switch	Data forwarding
Firewalls	Cisco ASA / Fortinet	Network security
Servers	Database & Application Servers	Banking services
UPS	Backup Power Supply	Prevent power failures

Transmission Media:

- **Single-Mode Fiber Optic Cable** – Long-distance communication
- **Multi-Mode Fiber Optic Cable (OM4)** – Data center communication
- Speed: **10 Gbps / 40 Gbps**
- Immune to electromagnetic interference (EMI)

Network Diagram:

```
  Router 1 -------- Router 2
     |\            /|
     | \          / |
     |  \        /  |
  Switch 1------Switch 2
     |\          /|
     | \        / |
Server 1------Server 2
```

Working:

1. All core routers, switches, and servers are connected through multiple links.
2. If one cable or device fails, traffic is automatically rerouted through another available path.
3. Routing protocols such as **OSPF** or **BGP** select the best path.
4. Fiber optic cables provide high-speed, reliable communication.
5. Redundant power supplies and UPS systems ensure uninterrupted network operation.

Advantages:

- High reliability
- No single point of failure
- Fast data transmission
- Automatic path redundancy
- Excellent network security
- Suitable for banking and financial institutions

Justification:

A **Partial Mesh Topology** is ideal for banks because it offers multiple communication paths, ensuring continuous network availability even if a device or link fails. High-speed **fiber optic cables**, redundant routers, Layer 3 switches, firewalls, and routing protocols like **OSPF/BGP** provide secure, fault-tolerant, and reliable communication required for financial transactions.

Conclusion:

A **Partial Mesh Topology** using **fiber optic cables**, **Layer 3 switches**, **enterprise routers**, and **firewalls** provides a secure, highly available, and fault-tolerant network that satisfies the banking industry's requirement for near-zero downtime and uninterrupted financial services.

Question 3: Design a Low-Cost Network for a Small Classroom Lab Using Bus Topology

Problem Analysis:

A small classroom laboratory requires a **low-cost** network for internet browsing and file sharing. Since the budget is limited, a **Bus Topology** is the most suitable option because all computers share a single communication cable, reducing installation costs.

Solution Design:

Topology Used
Bus Topology
Networking Devices

Device	Specification	Purpose
Computers	Desktop PCs	Student systems
Backbone Cable	RG-58 Coaxial Cable / Cat5e or Cat6	Data transmission
BNC T-Connector	50-ohm	Connect computers to backbone
Terminators	50-ohm	Prevent signal reflection
Teacher PC	Windows/Linux	File and printer sharing

Transmission Media:

- **RG-58 Coaxial Cable (50-ohm)** *(Traditional Bus Topology)*
- **Cat5e/Cat6 UTP Cable** *(Modern low-cost alternative)*
- Maximum cable length depends on the network design.

Network Diagram:

[T]---PC1---PC2---PC3---PC4---PC5---[T]
T = Terminator

Working:

1. All computers are connected to a single backbone cable.
2. When one computer sends data, the signal travels along the backbone.

3. The destination computer accepts the data, while other computers ignore it.
4. Terminators at both ends prevent signal reflection.
5. The teacher's PC can share files and printers with all students.

Advantages:

- Low installation cost
- Easy to install
- Requires less cable
- Suitable for small networks
- No dedicated server required

Limitations:

- A break in the backbone cable stops the entire network.
- Data collisions may occur when multiple devices transmit simultaneously.
- Performance decreases as more devices are added.
- Troubleshooting is difficult compared to star topology.

Justification:

A **Bus Topology** is the best choice for a small classroom with a limited budget because it uses a **single backbone cable**, minimizing hardware and cabling costs. It is suitable for light network usage such as internet browsing, file sharing, and classroom activities. Although it has limitations such as collisions and dependency on the backbone cable, it remains a cost-effective solution for small educational environments.

Conclusion:

A **Bus Topology** using **RG-58 coaxial cable**, **BNC connectors**, and **50-ohm terminators** provides an economical networking solution for a small classroom lab. It is easy to implement, requires minimal hardware, and is appropriate for environments with light network traffic.

Question 4: Design a Scalable Network for a Large University Campus Using Hybrid Topology

Problem Analysis:

A large university campus consists of multiple buildings such as classrooms, laboratories, libraries, hostels, and administration blocks. A single network topology cannot efficiently meet the requirements of scalability, reliability, and high-speed communication. Therefore, a **Hybrid Topology** is the best solution because it combines **Mesh/Ring** at the backbone and **Star Topology** within each building.

Solution Design:

Topology Used
Hybrid Topology (Mesh/Ring + Star)
Networking Devices

Device	**Specification**	**Purpose**
Core Router	Cisco ASR / Juniper MX	Campus backbone routing
Layer 3 Core Switch	Cisco Catalyst	Core network management
Distribution Switch	Layer 2/Layer 3	Connects each building
Access Switch	24/48-Port Switch	Connects end devices
Wireless Access Point	Wi-Fi 6	Wireless connectivity
End Devices	PCs, Laptops, Printers	User access

Transmission Media:

- **Single-Mode Fiber Optic Cable** – Backbone connection (10/40 Gbps)
- **Cat6 UTP Cable** – Building LAN connections (1 Gbps)
- **PoE (Power over Ethernet)** – Powers wireless access points

Network Diagram:

```
          Core Router
               |
       -------------------
       |                 |
  Core Switch------Core Switch
       |                 |
```

```
   -----------------------------
   |          |           |
Building A   Building B    Building C
    |          |          |
Access SW     Access SW     Access SW
||||      ||||     ||||
PCs/APs     PCs/APs     PCs/APs
```

Working:

1. Core routers and switches are connected using **fiber optic cables**.
2. Each building has a **distribution switch** connected to the campus backbone.
3. Access switches connect computers, printers, and Wi-Fi access points.
4. If one backbone link fails, traffic is rerouted through another path.
5. New buildings can be added easily without affecting the existing network.

Advantages:

- Highly scalable
- High-speed communication
- Reliable with redundant backbone links
- Easy network management
- Supports wired and wireless devices
- Easy expansion for future buildings

Justification:

A **Hybrid Topology** is ideal for a university campus because it combines the reliability of **Mesh/Ring Topology** in the backbone with the simplicity of **Star Topology** inside buildings. Fiber optic cables provide high-speed communication between buildings, while Cat6 cables connect end devices. This design ensures high availability, easy maintenance, and future scalability for thousands of users.

Conclusion:

A **Hybrid Topology** using **fiber optic backbone**, **Layer 3 core switches**, **distribution switches**, **access switches**, and **Cat6 UTP cables** provides a scalable, reliable, and high-performance network for a large university campus. It supports future expansion while ensuring efficient communication across all buildings.

Question 5: Design a Network for a Company with Multiple Departments Using VLANs

Problem Analysis:

A company with multiple departments such as **HR, Finance, Sales, and IT** requires a secure, scalable, and efficient network. The network should allow communication within each department while controlling communication between departments. A **Hierarchical Network Design** with **VLANs** is the best solution because it improves security, reduces broadcast traffic, and simplifies network management.

Solution Design:

Network Design
Hierarchical Topology with VLANs
Networking Devices

Device	**Specification**	**Purpose**
Layer 3 Switch	Cisco Catalyst	Inter-VLAN Routing
Layer 2 Switches	24-Port Managed Switch	Connect department devices
Router	Cisco ISR	Internet Connectivity
Firewall	Cisco ASA / Fortinet	Network Security
PCs & Printers	End Devices	Department Communication

VLAN Configuration:

Department	**VLAN ID**
HR	VLAN 10
Finance	VLAN 20
Sales	VLAN 30
IT	VLAN 40

Transmission Media:

- **Cat6 UTP Cable**
- Speed: **1 Gbps**
- Maximum Length: **100 meters**
- RJ-45 Connector

Network Diagram:

```
              Internet
                 |
            Router/Firewall
                 |
           Layer 3 Switch
      -------------------------------
      |      |      |        |
    VLAN10   VLAN20   VLAN30    VLAN40
     HR    Finance   Sales      IT
      |      |      |        |
   Layer2 SW Layer2 SW Layer2 SW Layer2 SW
      |      |      |        |
     PCs     PCs     PCs      PCs
```

Working:

1. Each department is assigned a separate **VLAN**.
2. Devices within the same VLAN communicate directly.
3. A **Layer 3 Switch** performs **Inter-VLAN Routing** to allow communication between departments when required.
4. The **Router** provides internet connectivity.
5. The **Firewall** protects the network from unauthorized access.
6. Redundant links can be added for higher reliability.

Advantages:

- Improved network security
- Reduced broadcast traffic
- Better network performance
- Easy management of departments
- Scalable for future expansion
- Reliable communication

Justification:

A **Hierarchical Network Design** with **VLANs** is ideal for a company because it logically separates departments, improves security, and reduces unnecessary network traffic. Layer 2 switches provide local communication, while a Layer 3 switch enables secure communication between VLANs. The router and firewall ensure safe internet access and protect the organization's network. This design is scalable, efficient, and widely used in modern enterprise environments.

Conclusion:

A **VLAN-based hierarchical network** using **Layer 2 switches**, a **Layer 3 switch**, **router**, **firewall**, and **Cat6 UTP cables** provides a secure, high-performance, and scalable solution for organizations with multiple departments. It enhances communication, simplifies management, and supports future business growth.